

HUM 444I PRESENTATION

Topic: Building an Ethical Framework for AI in Healthcare

Presenters

Group-13

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Introduction

Overview of AI in Healthcare:

Artificial Intelligence is transforming healthcare through advancements in clinical decisions, diagnosis and treatment. AI methodologies like machine learning, deep learning are vital for tasks like image classification, health outcome prediction and so on.

Importance of Ethics in AI and Healthcare:

AI integration into sensitive sectors like healthcare requires for a robust ethical framework. Ethical considerations are crucial to ensure safety, belief and responsible establishment of AI systems to prevent technological misuse.

Introduction

Global and Ethical Significance:

The ethical use of AI in healthcare is a global issue that impacts patient management, data breach and societal well-being. A unified ethical framework is required to guide AI practitioners and ensure safer AI for consumers worldwide.

Problem Statement

Definition of the Ethical Issue:

The vast embracement of AI in healthcare presents significant ethical challenges which include related to data privacy issues, algorithmic biasness, lack of transparency, accountability, patient safety and so on.

Affected Stakeholders:

The ethical challenges end up impacting a vast range of stakeholders including patients, physicians, healthcare professionals, AI developers and regulatory bodies.

Problem Statement

Example: Balancing Privacy with Technological Advancement:

A vital ethical dilemma is how to leverage AI's potential for innovation while protecting sensitive information and ensuring individual privacy. Our goal is to maximize the extraction of benefits from AI while keeping data safety as the primary priority without any compromise.

Objectives

Ethical Challenges regarding AI usage:

Identification and comprehension of major ethical principles influencing AI performance in healthcare, specifically in terms of data access level, algorithmic level and system levels.

Real-World Example:

Evaluating practical scenarios and instances where ethical issues in healthcare AI arise such as biased algorithms, data breaches, false diagnosis.

Recommendations for Ethical Practice:

Proposing an effective ethical framework and practical approach to design, construct, develop and initiate ethical AI in healthcare.

Literature Review

Related Studies and Ethical Guidelines:

Numerous studies highlighting critical ethical challenges in healthcare AI, including patient safety, data privacy, transparency, bias, explainability, sensitivity and accountability.

Professional Ethical Codes in AI & Healthcare

IEEE Code of Ethics

Human-Centered Design: Engineers must prioritize public safety, health and welfare in AI development and ensure whether AI systems are designed with the end-user in mind (IEEE, 2020).

Literature Review

IEEE Code of Ethics

Transparency & Accountability: Emphasizing the need for transparency in AI systems by requiring engineers to disclose potential risk factors and ensuring accountability in AI decision making.

Avoiding Harm: AI should not be used in ways that discriminate or violate privacy, aligning with the principle of non-maleficence.

Literature Review

ACM Code of Ethics

Fairness & Non-Discrimination: The ACM code stresses the importance of avoiding biases in AI systems, ensuring that healthcare applications do not reinforce existing inequalities (ACM, 2018).

Privacy Protection: Mandates strict safeguards for patient data, ensuring compliance with regulations like **HIPAA** and **GDPR**.

Responsible Use: AI should supplement human decisions, not alter it and must support the ethical principles.

Literature Review

NSPE Code of Ethics

Public Safety First: NSPE code emphasizes that engineers must ensure AI applications do not endanger patients, reinforcing non-maleficent principles (NSPE, 2019).

Competence & Integrity: Engineers need to maintain technical and ethical standards during AI development and ensure reliable actions.

Conflict of Interest: The code advises in opposition to the situations where commercial benefits tend to compromise patient well-being.

Literature Review

Ethical Concepts

Privacy: Protection of sensitive patient information and confidential documents

Algorithmic Bias: Systematic discrimination due to non-diverse datasets and ensuring equitable outcomes for all individuals.

Accountability: Identifying responsible stakeholders for AI system errors.

Literature Review

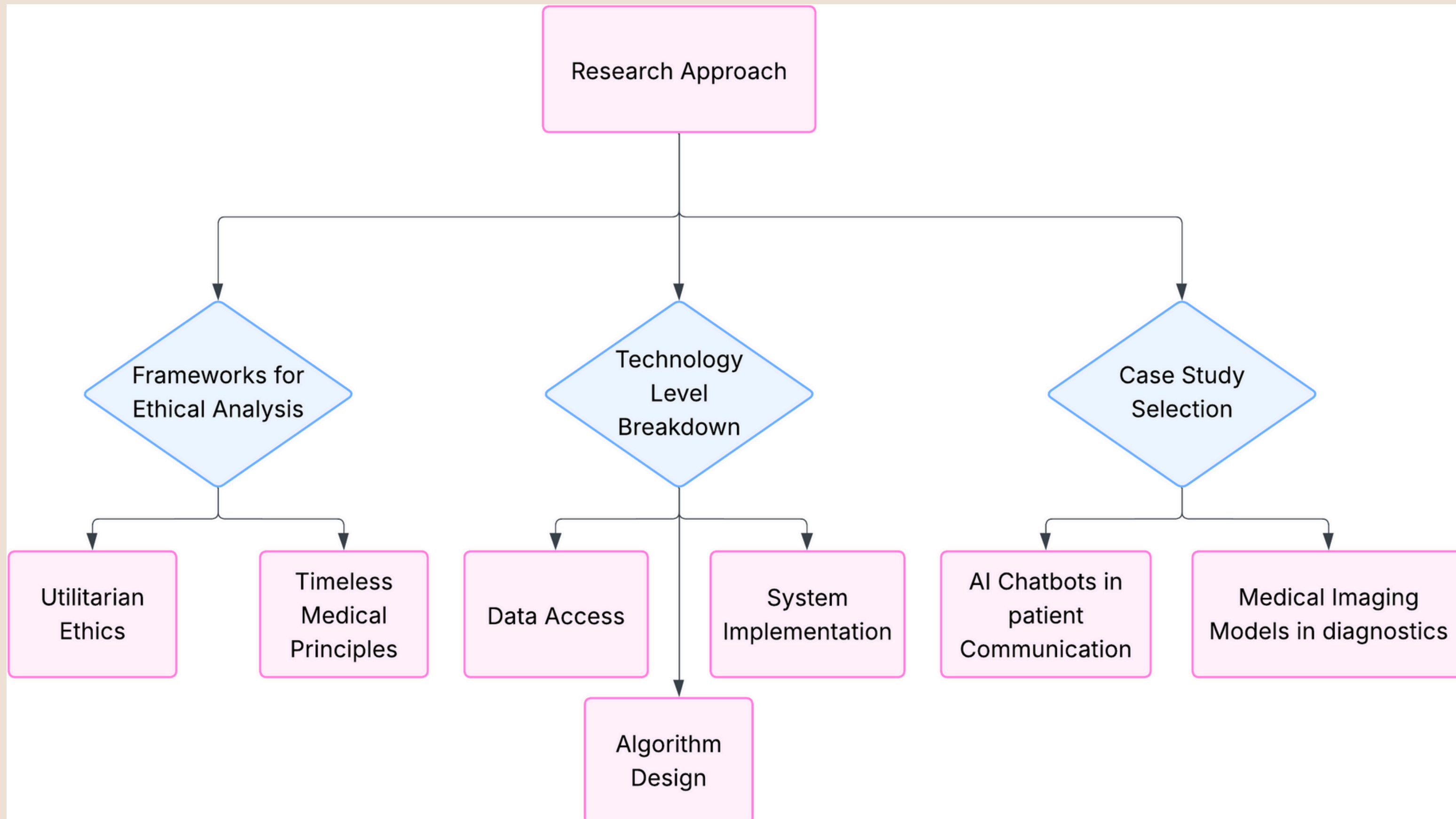
Ethical Concepts

Interpretability: Understanding the decision making process of AI systems.

Beneficence & Non-Maleficence: AI should be utilized to improve patient outcomes & Continuous monitoring of AI systems to prevent harm from false diagnoses or inappropriate recommendations.

Transparency: Fostering trust and allowing healthcare providers and patients to understand how AI systems arrive at their conclusions.

Methodology



Methodology

Research Approach

- Synthesized insights from a wide range of sources, including
 - In-depth **qualitative studies**,
 - Major reports on **expert opinions** (like those from the Pew Research Center),
 - And a thorough review of existing **ethical guidelines**..

Methodology

Frameworks for Ethical Analysis

- **Utilitarian Ethics**

- “What brings the greatest good to the most people?” This helped us weigh the benefits and risks of AI in healthcare, keeping patient well-being at the heart of our analysis.

- **Technology-Level Breakdown**

- Ethical concerns assessed at three levels:
 - **Data Access:** Who gets to use patient data, and how?
 - **Algorithm Design:** How do AI models make decisions, and are they fair?
 - **System Implementation:** How do these technologies fit into the bigger healthcare ecosystem

Methodology

Frameworks for Ethical Analysis

- **Timeless Medical Principles:** We connected modern AI risks back to the foundational principles of medical ethics, like "First, do no harm."

Case Study Selection

- Real-world applications explored:
 - **AI Chatbots** in patient communication
 - **Medical Imaging Models** in diagnostics

Case Study

AI in Action: Real-World Case Studies

Case Study 1:

Conversational Chatbots in Healthcare (GPT-4)

- **Use Cases:** To provide instant medical information, answer patient questions, and offer personalized health advice, especially when a doctor isn't immediately available.
- **Ethical Conflicts:**
 - **Hidden Biases:** Can the AI offer fair advice to everyone, or will it replicate harmful stereotypes from its training data?
 - **Privacy & Data Governance:** Who is "listening" to our most sensitive health concerns, and how is that data being protected?
 - **Overconfidence & Trust Decay:** Patients may over-rely on AI for life-or-death decisions or, conversely, lose trust in human doctors.
 - **Error Tolerance** A single piece of wrong advice isn't just a bug, it can lead to direct physical harm.

Case Study 1:

Foundational Models in Medical Imaging

- **Use Cases:** To help radiologists detect diseases like cancer earlier, analyze complex scans with greater accuracy, and create better treatment plans.
- **Ethical Conflicts:**
 - **Patient Confidentiality:** How do we train models on vast datasets of scans without risking the exposure of sensitive patient data?
 - **Bias in Diagnosis:** If an AI is trained primarily on one population, it may fail to accurately diagnose others, widening health disparities.
 - **Accountability Gaps:** When a complex AI system makes a mistake, who is responsible? The doctor, the hospital, or the developer?
 - **Lack of Transparency:** How can a clinician trust a recommendation if the AI can't explain why it made it?

Ethical Issues & Analysis

Main Ethical Issues in AI & Healthcare

- **Privacy & Security:** Risk of data breaches, unauthorized access, and misuse of sensitive patient information. Vulnerable to adversarial attacks and manipulation, threatening data integrity.
- **Fairness & Bias:** Non-diverse training data can lead to discrimination in diagnosis or treatment outcomes.
- **Accountability:** Difficult to trace responsibility in "black-box" models when errors or harms occur.
- **Informed Consent:** Patients may not fully understand how AI is involved in their care, weakening valid consent.
- **Regulation:** Our laws and regulations are struggling to keep pace with the rapid speed of technological innovation.

Ethical Issues & Analysis

Engineer's Role in Ethical Dilemmas

Engineers are the ethical architects of these systems. They must make their choice in

- **selecting data,**
- **designing algorithms,**
- **And deploying systems directly**

To make sure that AI is safe, fair, and trustworthy. Their mission is to balance innovation with a profound sense of responsibility.

Ethical Issues & Analysis

Ethical Frameworks & Viewpoints

To navigate these issues, we used three ethical lenses:

1. **The Greatest Good (Utilitarianism):** Prioritizes actions that maximize benefit and reduce harm for the greatest number.
2. **Our Core Duties (Deontology):** Emphasizes duty-based ethics respect for patient rights, autonomy, and privacy.
3. **The Moral Character (Virtue Ethics):** Focuses on the moral character of developers/users: encourages responsibility, integrity, and compassion in AI practice.

Solutions and Recommendations

Technical Solutions :

- **Glass-box FMs for openness :**
 - Glass-box models, as opposed to black-box models, are intended to be understandable and interpretable. encourages accountability, trust, and adherence to regulations.

Solutions and Recommendations

Technical Solutions :

- **Federated Learning to protect user privacy :**
 - AI models are trained across multiple sites (like hospitals) without sharing sensitive data, reducing breach risks and supporting compliance with HIPAA/GDPR.
- **Clinical validation and monitoring in real world :**
 - Protects privacy and addresses data imbalance; synthetic data boosts model accuracy without exposing real patient information.

Solutions and Recommendations

Policy Changes:

- **Regulatory frameworks for AI :** Establish dedicated agencies to oversee healthcare AI, set guidelines, approve models, and monitor their impact, similar to the FDA for drugs.
- **Updated informed consent:** Clearly define accountability for AI errors, whether developers, vendors, or hospitals to prioritize safety and protect patients.

Solutions and Recommendations

Policy Changes:

- **AI Liability Policies** : Require transparency in AI systems by enforcing reporting standards (like CONSORT-AI, SPIRIT-AI) and mandating model cards or data sheets.
- **Patient-Centric Consent Frameworks** : Update consent processes so patients can opt in or out anytime and are fully informed about AI's role, risks, and data use.

Solutions and Recommendations

Ethical System Design:

- **Privacy-by-Design** : Build privacy features like encryption and anonymization into AI systems from the start, not as afterthoughts.
- **Continuous Ethical Review** : Ensure ongoing ethical oversight throughout AI development, maintaining accountability as models change or update.

Role of Engineers

Core Responsibilities:

- **Ethical AI Development:** Engineers must embed ethics like accountability, safety, equity, and privacy into every stage of AI design, development, and deployment, not just as an afterthought.
- **Mitigate Algorithmic Bias:** Engineers must audit data, test for fairness, and apply debiasing methods at every stage to ensure AI treats all groups fairly and prevents harmful or unequal healthcare outcomes.
- **Promote Transparency and Disclosure of Limitations in AI Models :** Clearly communicate AI model limitations and uncertainties to users, ensuring safe, informed use and preventing over-reliance.

Role of Engineers

Ethical Guidelines :

- **Follow Professional Codes from IEEE, ACM, and National Engineering Bodies :** These groups offer well-established moral standards and conduct guidelines that direct responsible technology behavior.
- **Respect Patient Autonomy and Informed Consent in Technical Design:** Design AI systems to be transparent about their role, operation, and risks, ensuring patients are fully informed and can consent.
- **Ensure Transparency and Disclosure of Limitations in AI Models :** Openly disclose AI model limitations and uncertainties to support safe and informed use.

Role of Engineers

Innovation vs. Safety:

- **Avoid Over-Automation in Life-Critical Decisions** : AI should assist not replace human judgment in critical healthcare decisions to prevent automation bias.
- **Implement Fail-Safe Mechanisms and Fallback Protocols** : Build backup protocols like alerts and human handoffs for when AI is uncertain or out of scope, ensuring clinical safety.

Challenges in Implementation

Barriers :

- **Limited and Non-Representative Data Sources:** Biased or insufficient training data can lead to inaccurate or unfair AI results, risking health equity and reliability.
- **Black-Box Opacity:** The lack of interpretability in deep learning models makes it hard for clinicians to trust or understand AI decisions, impacting safety and accountability.

Challenges in Implementation

Barriers :

- **Infrastructure Gaps:** Hospitals in low-resource settings often lack the technology and expertise needed to deploy advanced AI, widening the digital divide in healthcare.
- **Lack of Experts in Ethical AI:** There's a lack of professionals skilled in both AI and medical ethics, limiting responsible development and oversight of healthcare AI.

Challenges in Implementation

Business vs Ethics :

- **Ethical Testing Is Undercut by Pressure to Commercialize Fast:**
Companies may skimp on ethical reviews in their haste to launch AI products, endangering patient safety and public confidence.
- **Corporate Secrecy Conflicts with Transparency Needs Explanation:**
Companies frequently protect their data and models as intellectual property, which restricts external auditing and accountability, particularly in delicate healthcare applications.
- **Ethical Vetting Viewed as a Barrier, Not a Priority :** A lot of businesses consider ethical review to be expensive and time-consuming, which results in little investment in bias testing, fairness, or patient rights protection.

Challenges in Implementation

Emerging Risks :

- **AI Hallucinations** : AI can generate confident but false information, risking patient safety through misdiagnosis or inappropriate treatment.
- **Bias in Automation Among Medical Professionals** : Clinicians may over-trust AI recommendations, which can undermine critical thinking and lead to dangerous errors if the AI is wrong.

Challenges in Implementation

Emerging Risks :

- **Adversarial Attacks:** Malicious input tweaks can trick AI models into making unsafe or incorrect decisions, posing risks to patient safety and clinical workflows.
- **Unclear Liability in AI-Driven Harm or Malpractice:** When AI causes harm, it's often unclear who is responsible—hospital, provider, or developer—creating legal uncertainty and weakening patient protections.

Future Directions

Cross-Disciplinary Collaboration :

- Foster partnerships between technologists, ethicists, healthcare workers, and legal experts.
- Encourage community involvement to ensure public values are reflected in AI.
- Promote co-design processes where patients and stakeholders shape AI solutions.
- Support interdisciplinary ethics committees for project approvals.

Future Directions

Focus Areas :

- **Ethical AI:** Create models that incorporate accountability, transparency, and fairness.
- **Cybersecurity:** Make an investment in cutting-edge defense for AI models and private medical data.
- **Rights of Patient Data:** Make sure that ownership, consent, and portability are all clearly defined.

Conclusion

- **Building Trustworthy AI in Healthcare:** Ethical design is the foundation for trust and successful adoption of AI in healthcare. Upholding principles like privacy, fairness, transparency, accountability, and strong regulation is essential.
- **Shared Ethical Responsibility:** AI developers, healthcare professionals, and all stakeholders have a duty to ensure that AI serves the best interests of patients protecting rights, safety, and well-being at every step.
- **A Call to Action:** Creating responsible AI in healthcare requires proactive efforts: combining technical innovation, thoughtful policy, and collaboration across disciplines to deliver safe, equitable, and trustworthy solutions for all.

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Thank you
for listening!